Medicines Company

By submitting this deck of case slides, the members of our team affirm that we all participated in the analysis of the case and the creation of this document.

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Qualitative Analysis

Value Proposition & Business Model: The company presents an intriguing concept: revitalizing drugs that may have previously failed to reach their full potential. However, this approach carries inherent risks, as there are often valid reasons for a drug's failure. The company essentially assumes they can achieve what others, often major pharmaceutical players, have not. Moreover, the public perception of this business model can be concerning. Medications are taken seriously because they can be life-saving, which might make customers hesitant to purchase a drug that had previously failed. Even if the drug's failure was due to reasons other than efficacy, there could be a potential stigma associated with it, making it challenging to market. Nevertheless, one major advantage of this model is the ability to bypass numerous early challenges in drug development. However, the Medicines Company forfeited this advantage when they acquired CTV-05, which was still in Phase I and ultimately led to a significant setback. To ensure success, the company should adhere to their original acquisition criteria, as outlined on Page 4. The Medicines Company's value proposition is twofold: it provides life-saving value for patients and economic value for healthcare professionals, hospitals, and investors. Their concept of rescuing drugs that would otherwise remain unused is transformative in the biomedical landscape. If they can acquire these drugs at a cost-effective price, it opens the door for numerous medications to enhance and potentially save lives. The value of a saved life is immeasurable, which suggests that the Medicines Company's drugs hold substantial value, both in terms of cost savings and profit potential. For example, their drug Angiomax demonstrated to physicians that it could save hospitals money and enhance their reputation by reducing complications. Nonetheless, it would be beneficial for the Medicines Company to refine their value proposition further by defining their target drug market explicitly. This could include focusing on medicines that outperform current alternatives or targeting populations with unmet medical needs.

Marketing and Sales: The Medicines Company astutely recognized three key stakeholders in their go-to-market strategy: doctors, pharmacists, and hospital administrators. Initially, the marketing of Angiomax with Dr. Plent primarily emphasized the cost savings for hospitals, an important aspect but not the only benefit the drug offered. Interestingly, Dr. Plent doesn't even mention the name "Angiomax" until the final sentence, suggesting that she might not be the ideal spokesperson for the drug. Fortunately, the company made a valuable addition by bringing in a Vice President of Sales and Marketing who identified various regions and outsourced some marketing efforts. They successfully targeted relevant stakeholders by establishing relationships with influential healthcare leaders in these regions through educational sessions. This regional approach was a wise move, recognizing the highly personal nature of healthcare. Patients often place a great deal of trust in their doctors and build strong relationships with them, which can go beyond their trust in large corporations. Moreover, having key opinion leaders in the medical field can significantly influence others to adopt the drug. Even before Angiomax received FDA approval, the Medicines Company's marketing strategy was laying the groundwork for its introduction by raising questions about the viability of heparin, with the goal of supplanting heparin and elevating Angiomax in the medical community. The company effectively disseminated this message, including through the publication of articles in the *Journal of Invasive Cardiology*.

Market Competitiveness: The pharmaceutical market, as a whole, is vast, with major players generating billions of dollars in revenue. More specifically, the market for blood thinners, especially those used in angioplasty procedures, is substantial and firmly established, with heparin serving as the standard treatment for over 85 years. With the aging population in the United States on the rise, Angiomax becomes an increasingly relevant treatment option for older adults who are at higher risk of cardiovascular diseases. The Medicines Company can tap into this demographic shift by targeting hospitals and healthcare providers that primarily serve the aging population. However, the expected growth in generic blood thinner sales, as several blockbuster drugs go off-patent, may pose a challenge for Angiomax. Generic alternatives could potentially enter the market at lower price points. To effectively navigate this competition, the Medicines Company needs to emphasize the unique attributes, advantages, and superior efficacy of Angiomax in comparison to generic blood thinners like heparin, thereby justifying its higher cost. Heparin has a long and established history of safe and effective use, but being a generic drug, it faces competition from multiple manufacturers, resulting in market price pressures. To succeed in the market, the Medicines Company should focus on hospitals at the forefront of interventional cardiology that are open to adopting new treatments and technologies. This could include leading academic medical centers, specialized heart centers, and hospitals with experienced interventional cardiologists willing to transition from heparin to Angiomax based on available data and clinical trial results.

Quantitative Analysis

HIGH RISK - 40% of all angioplasties			
Outcome within 7 days of treatment	**Heparin**	**Angiomax**	**Difference**
(# of patients in condition)	2,151	2,161	
Death	0.20%	0.20%	0.00%
Heart Attack	4.20%	3.30%	0.90%
Need for a Repeat Angioplasty	2.80%	2.50%	0.30%
Experienced Major Bleeding	9.30%	3.50%	5.80%
Number of angioplasties:	700,000		
Number of high risk angioplasties:	280,000		
Cost per angioplasty:	$9,500.00		
Cost per additional treatment:	$8,000.00		
Average number of additional treatments including death with Heparin:	46,200		
Average number of additional treatments including death with Angiomax:	26,600		
Difference:	19,600		
Savings:	$156,800,000.00		

VERY HIGH RISK - 10% of all angioplasties			
Outcome within 7 days of treatment	**Heparin**	**Angiomax**	**Difference**
(# of patients in condition)	372	369	
Death	0.50%	0.00%	0.50%
Heart Attack	5.60%	3.00%	2.60%
Need for a Repeat Angioplasty	3.50%	2.40%	1.10%
Experienced Major Bleeding	11.80%	2.40%	9.40%
Number of angioplasties:	700,000		
Number of VERY high risk angioplasties:	70,000		
Cost per angioplasty:	$9,500.00		
Cost per additional treatment:	$8,000.00		
Average number of additional treatments including death with Heparin:	14,980		
Average number of additional treatments including death with Angiomax:	5,460		
Difference:	9,520		
Savings:			$76,160,000.00

92%	700,000 angioplasties conducted annually	644,000		
	Price of Angiomax			
% of adopters from the targeted medical centers	**$200.00**	**$250.00**	**$350.00**	**$500.00**
20%	$25,760,000.00	$32,200,000.00	$45,080,000.00	$64,400,000.00
40%	$51,520,000.00	$64,400,000.00	$90,160,000.00	$128,800,000.00
60%	$77,280,000.00	$96,600,000.00	$135,240,000.00	$193,200,000.00
80%	$103,040,000.00	$128,800,000.00	$180,320,000.00	$257,600,000.00
100%	$128,800,000.00	$161,000,000.00	$225,400,000.00	$322,000,000.00

Hospitals face an additional cost of $8,000 when providing treatment to individuals who have suffered a heart attack, need a repeat angioplasty, or have significant bleeding. Hospitals are often obligated to absorb these extra costs. By examining the data in Table A and Table B from the case, it becomes evident that Angiomax not only enhances patient survival but also results in cost savings for hospitals by reducing the need for repetitive procedures and extended hospitalizations due to complications. By analyzing the percentage change in complication rates when comparing Heparin to Angiomax, it is clear that 19,600 fewer patients require additional medical interventions. This leads to a reduction of $156 million in hospital expenditures for these patients. When this same analysis is extended to Very High-Risk procedures (10% of angioplasties), hospitals have a cost saving of $76 million on patient care. Hospitals also enhance their reputation by improving their success rates, rendering Angiomax an extremely marketable product. These calculations show that Angiomax provides significantly higher value-in-use than Heparin, with variations in value-in-use contingent on the patient's risk level.

Annually, 700,000 Balloon Angioplasties are performed, with 92% of these procedures carried out by clinics, making The Medicine Company's target 644,000 angioplasties. Taking into account various adoption rates among medical centers and potential pricing options, the table illustrates the range of potential annual revenues from Angiomax sales. By setting the drug's price at $250, hospitals can have significant savings while still allowing The Medicine Company to have substantial profits. When approximately 60% of the targeted medical centers embrace this pricing strategy, they will achieve their objective of reaching $100 million in annual revenue.

Recommendation

Marketing Recommendations:

- **Craft a more holistic narrative around Angiomax and future drugs:** TMC's initial marketing approach overemphasized the cost-effectiveness of Angiomax without highlighting other advantages such as efficacy. The company should emphasize a broader range of selling points to better demonstrate the drug's value proposition to different stakeholders in healthcare. This will help assuage concerns about TMC's business model of "rescuing" previously failed drugs and provide customers with more reasons to adopt their drugs.

- **Price Angiomax at $250:** After testing potential prices from $200 to $500, we propose an optimal price of $250. This is based on the cost to produce, value (hospitals will save over $200 million), projected clinic adoption rates (20%-100%), and an annual revenue target of $100 million. We also accounted for potential reluctance to switch to a more expensive drug, thus pricing Angiomax at the lower end of the tested range.

- **Prioritize sales efforts towards well-funded, reputable hospitals:** Since new drugs with high revenue potential for The Medicines Company will be priced at a premium, they should target hospitals with the financial capacity to transition to more expensive drugs. Moreover, by selling to renowned academic hospitals like Mt. Sinai or Johns Hopkins, these drugs can become an industry norm for community/smaller hospitals to follow.

- **Market to physicians that use competitor Heparin for treatments beyond angioplasties:** The Medicines Company is currently only reaching 20% of 3.5 million annual Heparin users. Raising awareness about Angiomax in doctors and clinics that currently use Heparin for other indications will help TMC reach the other 80% of Heparin users.

Business Model Recommendations:

- **Prioritize successfully launching Angiomax:** Even the approval of one drug is strong evidence in favor of The Medicines Company's controversial business model. This will enhance the company's standing among competitors, and thus, brand recognition. Furthermore, it will increase customer retention by gaining trust in providers (trust is important as they face high stakes in prescribing treatment.)

- **Limit future acquisitions to Phase III drugs:** This will limit unnecessary development costs, maximize chances of success in clinical trials, and expedite market entry. Setbacks and delays in the drug development pipeline are both expensive and risky for the company's reputation.

- **Continue R&D efforts for CTV-05 despite its extended timeline:** CTV-05 has shown enough promise in P2 trials to warrant continuation in the pipeline. TMC should plan for market entry 3-7 years after CTV-05 completes P2 trials.